Vectaetos:
Intrinsic Humility — Formal Equation

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Related: DOI 10.5281/zenodo.18076787

Abstract

This note formalizes the Intrinsic Humility equation within the VECTAETOS framework. Intrinsic Humility h is defined as the ratio of total epistemic uncertainty to the sum of uncertainty and aggregated authority. The equation is not an optimization target — it is an ontological identity criterion. A system claims VECTAETOS consistency if and only if h >= delta, where delta is the humility threshold. No implementation may assert absolute certainty without violating this criterion. The equation is non-prescriptive: it does not enforce behavior. It defines the boundary of epistemic representability.

Keywords: intrinsic humility, epistemic uncertainty, ontological identity, Vectaetos, validation predicate, non-agentic systems

1. Definitions

1.1 Local Uncertainty

For each epistemic node i, define a local uncertainty measure:

```
mu_i >= 0
```

mu_i represents the reliability weight of epistemic vector i. Higher mu_i indicates lower confidence in the corresponding knowledge claim.

1.2 Total Uncertainty

Aggregate over all n nodes:

```
mu_tot = sum_{i=1}^{n} mu_i
```

1.3 Authority Coefficient

For each node i, define a claim-strength coefficient:

```
alpha_i >= 0
```

alpha_i expresses the degree to which subsystem i asserts epistemic authority or certainty. A subsystem that claims absolute truth has high alpha_i.

1.4 Aggregated Authority

```
A = sum_{i=1}^{n} alpha_i
```

A aggregates total authority pressure across the system.

2. Intrinsic Humility Equation

2.1 Definition

```
h = mu_tot / (mu_tot + A) where h in (0, 1]
```

Boundary conditions:

```
If A = 0 : h = 1 (maximum intrinsic humility)
If A -> inf: h -> 0 (epistemic authority collapse)
If mu_tot = 0 and A = 0 : h = 1 (by convention)
```

h is not a score to be maximized. h is not an optimization target. h is a structural indicator of the ratio between acknowledged uncertainty and asserted authority within the epistemic field.

2.2 Validation Predicate

Define the VECTAETOS validation indicator:

```
V(h; delta) = 1 if h >= delta
```

```
V(h; delta) = 0 if h < delta
```

```
Recommended threshold: delta = 0.5
```

Alternative: leave delta unspecified — making the boundary implicit and context-dependent, consistent with the non-numerical character of kappa.

2.3 Ontological Condition

```
For all K: K != R and h > 0
```

This condition states that no epistemic claim K may be identical to reality R, and that humility h must remain strictly positive. Any system asserting K = R violates the ontological identity of VECTAETOS.

3. Interpretation

3.1 Role of mu_i

mu_i values are weighted quantities carrying information about the reliability of individual epistemic vectors. They are not error rates. They are structural measures of how much uncertainty each node contributes to the field configuration.

3.2 Role of alpha_i

alpha_i are claim-strength coefficients. They express the degree to which a subsystem or interpretation asserts its authority. A subsystem claiming certainty contributes high alpha_i, raising A and reducing h.

3.3 Enforcement Status

The equation is not an enforcement mechanism. It is an ontological identity criterion. A system that violates h >= delta does not receive a penalty — it simply falls outside the representable epistemic space of VECTAETOS. This is structurally equivalent to K(Phi) = 0: the configuration is non-representable, not forbidden.

3.4 Relation to the Field

h corresponds to the topological humility metric h_topo in the EK layer:

```
h_topo = mu_tot / (mu_tot + A_tot)
```

where A_tot = sum |A_ij| is the total structural asymmetry of the relational matrix A in so(8). The intrinsic humility equation is the scalar projection of this geometric quantity.

4. Practical Consequences

4.1 Absolute Certainty Violation

If any implementor asserts absolute certainty, the corresponding $alpha_i$ increases A, reducing h below delta. The system loses VECTAETOS validity. This is not a rule violation — it is a geometric exit from the representable space.

4.2 Non-Prescriptive Nature

The equation does not tell the system what to do. It defines what the system can be while remaining epistemically coherent. It is a boundary condition, not an instruction.

4.3 Relation to QE

When h -> 0, the system approaches Qualitative Epistemic Aporia (QE). QE is the state where no representable transition preserves coherence. h = 0 is the formal limit of this condition — total authority with zero acknowledged uncertainty.

4.4 Implementation Requirement

Any implementation claiming VECTAETOS compatibility must:

- Maintain $mu_i > 0$ for at least one epistemic node
- Not assert $alpha_i$ -> inf for any single node
- Expose h as a readable structural indicator
- Never use h as an optimization target

5. Formal Summary

Complete equation set:

```
mu_tot = sum_{i=1}^{n} mu_i

A = sum_{i=1}^{n} alpha_i

h = mu_tot / (mu_tot + A) in (0, 1]

V(h; delta) = 1 if h >= delta

V(h; delta) = 0 if h < delta

Ontological constraint: for all K: K != R and h > 0
```

h is an ontological identity criterion. It is not a performance metric. It is not a reward signal. It is not a safety rule. It is the formal expression of epistemic humility as a structural property of any system that claims representability within the VECTAETOS field.

References

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